Ziyi Xu

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Research Interests

My research is driven by one central question: **how can the physical and statistical structure of wireless environments be exploited to develop accurate, adaptive, and efficient communication and sensing systems?** I pursue three connected directions: **near-field and XL-MIMO**, leveraging Fresnel-domain representations and propagation characteristics for channel estimation and low-overhead beam training; **distribution-aware wireless learning and inference**, developing provable meta-learning algorithms, scalable generative channel priors, and score-guided information-theoretic sensing design; and **environment-aware ISAC**, exploiting target, clutter, and communication information for joint transmit–receive optimization. Across these directions, I combine physical modeling, signal processing, information theory, optimization, and deep learning to bridge theoretical foundations and practical wireless system design.

Education

2022–Present **Ph.D. Candidate, Information Engineering**, Chinese University of Hong Kong.
Expected thesis submission date: Dec 2026.
Supervisor: **Prof. Ying-Jun Angela Zhang**

2018–2022 **B.Sc., Electronic and Information Engineering**, Harbin Institute of Technology
Top Ten Outstanding Graduate, School of Honor.

Publications

* *Corresponding author;* † *Equal contribution.*

Near-Field Channel Modeling and Representations

[J1] **Optics-Inspired Foundations for Near-Field Wireless Communications**
Ziyi Xu†, Lin Chen†, Ran Li†, Ying-Jun Angela Zhang
npj Wireless Technology. vol. 2, art. no. 56, Aug. 2026

[J2] **Near Field Sparse Representation and Dictionary Design Using Discrete Fresnel Transform**
Ziyi Xu, Shuoyao Wang, Ying-Jun Angela Zhang
IEEE Transactions on Wireless Communications. vol. 25, pp. 6760–6773, 2026

[C1] **Near Field Dictionary Design and Channel Estimation Using Discrete Fresnel Transform**
Ziyi Xu, Shuoyao Wang, Ying-Jun Angela Zhang
58th Asilomar Conference on Signals, Systems, and Computers. pp. 1690–1694, 2024

Efficient Beam Training for Near-Field and XL-MIMO Systems

[J3] **3D Near-Field Beam Training for Uniform Planar Arrays: Low-Overhead Approaches Leveraging Beam Diverging**
Ran Li, **Ziyi Xu***, Ying-Jun Angela Zhang
IEEE Transactions on Wireless Communications. vol. 25, pp. 22941–22954, 2026

[C2] **Divergence Meets Dispersion: Efficient Wideband Beam Training for XL-MIMO**
Ran Li, **Ziyi Xu**, Wei Guo, Ying-Jun Angela Zhang
IEEE Wireless Communications and Networking Conference (WCNC). pp. 1–6, 2026.
BEST PAPER AWARD

[C3] **3D Near-Field Beam Training for Uniform Planar Arrays through Beam Diverging**
Ran Li, **Ziyi Xu**, Ying-Jun Angela Zhang
IEEE Global Communications Conference (GLOBECOM). pp. 1501–1506, 2025

[C4] **Near-Field Beam Training Through Beam Diverging**
Ran Li, **Ziyi Xu**, Ying-Jun Angela Zhang
IEEE International Conference on Communications (ICC). pp. 1718–1723, 2025

Adaptive Learning for Wireless Communications

[J4] **Scenario-Adaptive Meta-Learning for mmWave Beam Alignment**
Ziyi Xu, Shuoyao Wang, Ying-Jun Angela Zhang
IEEE Transactions on Wireless Communications. vol. 24, no. 4, pp. 3192–3208, 2025

[C5] **Few-Pilot Channel Estimation via Tuning-Free MAML**

Ziyi Xu, Zhiwei Zhai, Wenjing Yan, Ying-Jun Angela Zhang
IEEE Global Communications Conference Workshops (GLOBECOM Workshops). Accepted, 2026

Distribution-Aware Sensing and Generative Inference

[C6] **Channel Collage: Scalable Diffusion-Based Prior Learning for ELAA Channels**
Ziyi Xu, Chang Cai, Shuoyao Wang, Ying-Jun Angela Zhang
Asilomar Conference on Signals, Systems, and Computers. Accepted, 2026

[C7] **On Mutual Information Maximization in Linear Sensing with Arbitrary Channel Priors**
Ziyi Xu, Shuoyao Wang, Ying-Jun Angela Zhang
IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM). pp. 1–4, 2026

Selected Ongoing Research

[W1] **From Physics-Informed Channel Models to Generative Wireless Inference**
Building on Fresnel-domain sparse channel representations (*IEEE TWC 2026*) and compositional diffusion priors (*Asilomar 2026*), developing scalable, structure-aware inference for extremely large antenna arrays. Current directions include score-aided turbo message passing for compressed channel recovery and mutual-information-driven sensing matrix design using learned channel priors.

[W2] **Low-Overhead Beam Training and Environment-Aware ISAC**
Extending prior work on near-field beam-diverging and wideband beam training (*IEEE TWC/WCNC 2026*) toward clutter-aware ISAC. Developing sparse-periodic DFT sounding to construct compact beam-level representations of communication links, targets, and clutter, enabling joint beam, power, and receive combining optimization for sensing SCNR under communication SINR constraints.

[W3] **Scenario-Adaptive Wireless Learning with Provable Guarantees**
Building on scenario-adaptive mmWave beam alignment (*IEEE TWC 2025*) and few-pilot channel estimation (*GLOBECOM Workshops 2026*), developing tuning-free MAML and Hessian-free variants with normalized-momentum updates. The framework addresses biased meta-gradients without unknown problem constants and achieves optimal $\mathcal{O}(\epsilon^{-4})$ nonconvex complexity. *Theory manuscript under submission.*

Research Experience

2022–Present **Doctoral Research**, The Chinese University of Hong Kong
Supervisor: **Prof. Ying-Jun Angela Zhang**
- Developed physics-aware and learning-enabled methods for near-field/mmWave beam training, Fresnel-domain sparse representation, channel estimation, and adaptive beam management for 6G XL-MIMO.

Mar–Jun 2026 **Visiting Ph.D. Student**, Communication Systems Division, Linköping University
Host supervisors: **Erik G. Larsson, Karl-Ludwig Besser, and Diana P. Moya Osorio**
- Investigated clutter-aware ISAC transceiver design, combining beam sounding, power allocation, and receive combining under communication QoS constraints.

Oct 2021–Jan 2022 **Research Assistant**, University of Hong Kong
Research supervisor: **Prof. Chenshu Wu**
- Built a Wi-Fi CSI pipeline for indoor speed estimation using Intel 5300 hardware, 57 subcarriers (20 MHz), denoising, and Doppler spectral analysis.

Jul 2019–Jul 2020 **Research Project**, Harbin Institute of Technology
Project supervisor: **Prof. Hongzhi Li**
- Co-developed an STM32-controlled badminton serving machine and a Raspberry Pi vision server for Hough-circle shot detection.
- Integrated mobile-app cloud control; demo: video link.

Honors & Awards

2026 **Best Paper Award**, IEEE WCNC, for *Divergence Meets Dispersion: Efficient Wideband Beam Training for XL-MIMO.*

2023 **Second Prize**, inaugural IEEE ComSoc Three Minute Thesis (3MT) Competition.

2022 **Outstanding Tutor Award**, The Chinese University of Hong Kong.

2020 **National Scholarship**, Harbin Institute of Technology (2019–2020 academic year).

2019 **National Scholarship**, Harbin Institute of Technology (2018–2019 academic year).